Box 47.1 Examples of vascular lesions producing renal hypoperfusion and the syndrome of renovascular hypertension

Unilateral disease (analogous to 1-clip-2-kidney hypertension)

Unilateral atherosclerotic renal artery stenosis
Unilateral fibromuscular dysplasia (FMD)[200]
  Medial fibroplasia
  Perimedial fibroplasia
  Intimal fibroplasia
  Medial hyperplasia
Renal artery aneurysm
Arterial embolus
Arteriovenous fistula (congenital/traumatic)
Segmental arterial occlusion/dissection (posttraumatic)
Extrinsic compression of renal artery (e.g., pheochromocytoma)
Renal compression (e.g., metastatic tumor)

Bilateral disease or solitary functioning kidney (analogous to 1-clip-1-kidney model)

Stenosis to a solitary functioning kidney
Bilateral renal arterial stenosis
Aortic coarctation/mid-aortic syndrome (sometimes seen in children)
Systemic vasculitis (e.g., Takayasu and polyarteritis)
Atheroembolic disease
Vascular occlusion due to endovascular aortic stent graft

Box_47_1

Box 47.2 Clinical features of patients with renovascular hypertension

Syndromes associated with renovascular hypertension

1. Early- or late-onset hypertension (<30 years >50 years)
2. Acceleration of treated essential hypertension
3. Deterioration of renal function in treated essential hypertension
4. Acute renal failure during treatment of hypertension
5. "Flash" pulmonary edema
6. Progressive renal failure
7. Refractory congestive cardiac failure

The above "syndromes" should alert the clinician to the possible contribution of renovascular disease in a given patient. Numbers 5 to 7 are most common in patients with bilateral disease, many of whom are treated as "essential hypertension" until these characteristics appear.

Box_47_2

Box 47.3 Renovascular disease and ischemic nephropathy

Goals of diagnostic evaluation

- Establish presence of renal artery stenosis: location and type of lesion
- Establish whether unilateral or bilateral stenosis (or stenosis to a solitary kidney) is present
- Establish presence and function of stenotic and nonstenotic kidneys
- Establish hemodynamic severity of renal arterial disease
- Evaluate progression of vascular occlusion/renal atrophy
- Plan vascular intervention: degree and location of atherosclerotic disease

Goals of therapy

I. Improved blood pressure control:
  Prevent morbidity and mortality of high blood pressure
  Improve blood pressure control and reduce medication requirement
II. Preservation of renal function
  Reduce risk of renal adverse perfusion from use of antihypertensive agents
  Reduce episodes of circulatory congestion ("flash" pulmonary edema)
  Reduce risk of progressive vascular occlusion causing loss of renal function: "preservation of renal function"
  Salvage renal function (i.e., recover glomerular filtration rate)
III. Recovery of volume regulation and fluid excretion
  Reduce circulatory overload and diuretic resistance
  Reduce episodes of "flash pulmonary edema"

Box_47_3

Box 47.4 Complications after percutaneous renal angioplasty and stenting of the renal arteries

Minor (most frequently reported)

Groin hematoma
Puncture site trauma

Major (reported in 71/799 treated arteries (9%)[186]

Hemorrhage requiring transfusion
Femoral artery pseudoaneurysm needing repair
Brachial artery traumatic injury needing repair
Renal artery perforation leading to surgical intervention
Stent thrombosis: surgical or antithrombotic intervention
Distal renal artery embolus
Iliac artery dissection
Segmental renal infarction
Cholesterol embolism: renal
Peripheral atheroemboli
Aortic dissection[168]
Restenosis: 16% (range: 0%–39%)
Deterioration of renal function: 26% (range: 0%–45%)
Combined complication rate over 24 months: stenting after failed PTRA: 24% (222)
Mortality attributed to procedure: 0.5%
Procedure-related complications: 51/379 patients in 10 series: 13.5%
Procedure complications reported from CORAL angiographic core lab (2014): (159)
  Dissection 11/495 (2.2%)
  Branch occlusion 6/495 (1.2%)
  Distal embolization 6/495 (1.2%)
  Single events: wire perforation, vessel rupture, pseudoaneurysm
  Total events; 26/495 (5.2%)

Box_47_4

Box 47.5 Surgical procedures applied to reconstruction of the renal artery and/or reversal of renovascular hypertension

Ablative surgery: removal of a "pressor" kidney
  Nephrectomy: direct or laparoscopic
  Partial nephrectomy
  Renal artery reconstruction (require aortic approach)
  Renal endarterectomy
  Transaortic endarterectomy
  Resection and reanastomosis: suitable for focal lesions
  Aortorenal bypass graft
  Extraanatomic procedures: (may avoid direct manipulation of the aorta)
  Require adequate alternate circulation without stenosis at celiac origin
  Splenorenal bypass graft
  Hepatorenal bypass graft
  Autotransplantation with ex-vivo reconstruction
  Gastroduodenal, superior mesenteric, iliac-to-renal bypass grafts

Modified from Libertino.[207,208]

Box_47_5

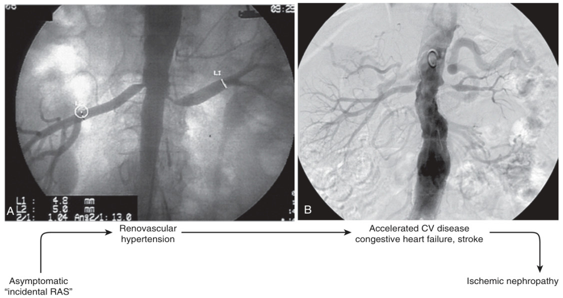


Fig. 47.1 Spectrum of atherosclerotic RVD (ARVD). Panal A depicts an aortogram obtained during coronary angiography demonstrating moderate "incidental" stenosis of both renal arteries in a 67-year-old man with symptomatic coronary disease. Panal B illustrates more severe occlusive disease observed in a 68-year-old woman presenting with severe hypertension and episodes of "flash pulmonary edema." RVD commonly develops in the setting of atherosclerotic disease elsewhere and may be associated with multiple clinical syndromes ranging from renovascular hypertension to accelerated cardiovascular decompensation and ischemic nephropathy. *RAS,* Renal artery stenosis. (Modified from Herrmann SM, Saad A, Textor SC. Management of atherosclerotic renovascular disease after Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL). *Nephrol Dial Transplant*. 2015;30(3):366–375 with permission.)

Fig_47_1

Fig. 47.2 (A) Measured fall in arterial pressure and blood flow across stenotic lesions induced in experimental animals. The degree of stenosis was determined using latex casts after completion of the experiment. These data indicate that "critical" lesions require 70% to 80% luminal obstruction before hemodynamic effects can be detected. (B) Effects of a balloon-induced, unilateral, controlled, graded stenosis (expressed as Pd/Pa ratio) on plasma renin concentration in the aorta (*green*), in the vein of the stenotic kidney (*blue*), and in the vein of the non-stenotic kidney (*red*). P_a, Aortic pressure; P_d, distal pressure. (A from May AG, Van de Berg L, DeWeese JA, et al. Critical arterial stenosis. *Surgery*. 1963;54:250–259; B from De Bruyne B, Manoharan G, Pijls NHJ, et al. Assessment of renal artery stenosis severity by pressure gradient measurements. *J Am Coll Cardiol.* 2006;48:1851–1855.)

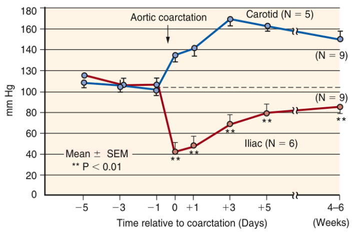


Fig. 47.3 Systemic arterial pressure (carotid) and poststenotic renal perfusion pressures (iliac) in an aortic coarctation model with a clip placed between the right and left renal arteries. Measurements were obtained in conscious animals during development of renovascular hypertension. They illustrate the fact that despite a persistent gradient across the stenosis, renal perfusion pressure (inferred from Iliac pressure) returns to near normal levels as a result of systemic hypertension. A corollary of this pressure gradient is that reduction of systemic pressures lowers poststenotic perfusion pressures, sometimes below the range of autoregulation. (From Textor SC, Smith-Powell L. Post-stenotic arterial pressures, renal haemodynamics and sodium excretion during graded pressure reduction in conscious rats with one- and two-kidney coarctation hypertension. *J Hypertens*. 1988;6(4):311–319 with permission.)

Fig_47_2

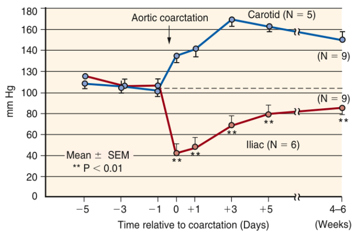


Fig. 47.3 Systemic arterial pressure (carotid) and poststenotic renal perfusion pressures (iliac) in an aortic coarctation model with a clip placed between the right and left renal arteries. Measurements were obtained in conscious animals during development of renovascular hypertension. They illustrate the fact that despite a persistent gradient across the stenosis, renal perfusion pressure (inferred from Iliac pressure) returns to near normal levels as a result of systemic hypertension. A corollary of this pressure gradient is that reduction of systemic pressures lowers poststenotic perfusion pressures, sometimes below the range of autoregulation. (From Textor SC, Smith-Powell L. Post-stenotic arterial pressures, renal haemodynamics and sodium excretion during graded pressure reduction in conscious rats with one- and two-kidney coarctation hypertension. *J Hypertens*. 1988;6(4):311–319 with permission.)

Fig_47_3

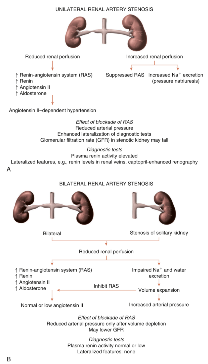


Fig. 47.4 Schematic view of two-kidney (A) and one-kidney (B) renovascular hypertension. These models differ by the presence of a contralateral kidney exposed to elevated perfusion pressures in two-kidney hypertension. The nonstenotic kidney tends to allow pressure natriuresis to ensue and produces ongoing stimulation of renin release from the stenotic kidney. The one-kidney model eventually produces sodium retention and a fall in renin with minimal evidence of angiotensin dependence unless sodium depletion is achieved.

Fig_47_4

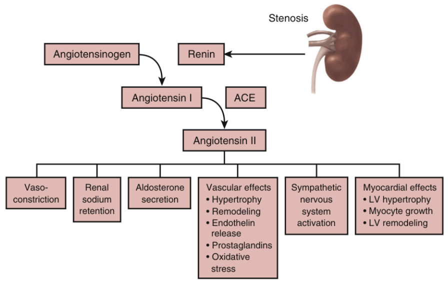


Fig. 47.5 Schematic view of activation of the renin-angiotensin system beyond a renal artery stenotic lesion. Generation of circulating and local angiotensin II leads to widespread effects including sodium retention, efferent arteriolar vasoconstriction, and elevated systemic vascular resistance. Additional studies implicate angiotensin II in many other pathways of vascular and cardiac smooth muscle remodeling, activation of inflammatory and fibrogenic cytokines, coagulation factors, and induction of other vasoactive systems. *ACE,* Angiotensin-converting enzyme.

Fig_47_5

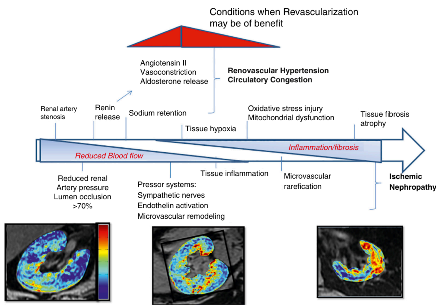


Fig. 47.6 Progression of renovascular disease and the limitations of revascularization. *Central horizontal arrow* depicts the sequence of events associated with progressive renovascular occlusion. Reductions in blood flow and perfusion pressure develop only after substantial (>70%) lumen occlusion, leading to activation of the renin-angiotensin system. Bottom images depict blood-oxygen-level-dependent (BOLD) MR slices mapping deoxyhemoglobin within the kidney. Despite reduced blood flow, renal oxygenation is significantly reduced (bottom right panel) only with severe and prolonged occlusion, which is associated with activation of inflammatory injury, vascular rarefication, and tissue fibrosis. Renal revascularization can restore kidney perfusion and reverse these processes only under conditions that have not become irreversible (depicted in red). (Reprinted with permission from Bhalla V, Textor SC, Beckman JA, et al. Revascularization for renovascular disease: a scientific statement from the American Heart Association. *Hypertension*. 2022;79(8):e128–e143. © 2022 American Heart Association, Inc.)

Fig_47_6

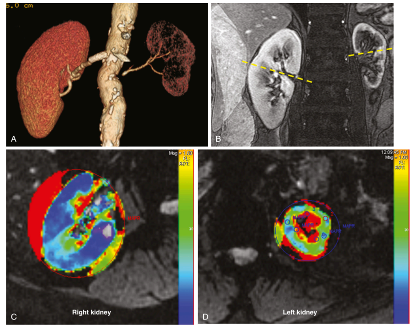


Fig. 47.7 Computed tomography angiogram (A) of high-grade vascular occlusion leading to left kidney atrophy while the right kidney is well perfused beyond an endovascular stent. T2 signals from magnetic resonance imaging (B) illustrate a gradient of perfusion from cortex to lower levels in the deeper medullary sections, especially in the left kidney. Mapping of the blood-oxygen-level-dependent (BOLD) magnetic resonance R2* levels (a function of deoxyhemoglobin) (C and D) illustrates the normal gradient in the right kidney from cortex (low R2*, blue) to deeper medullary segments that have progressively higher levels of R2* (green to red). The R2* map of the left kidney (D) illustrates overt cortical hypoxia evident beyond critical vascular occlusion associated with a larger fraction of the slice having high R2* values (red) associated with fractional medullary hypoxia.

Fig_47_7

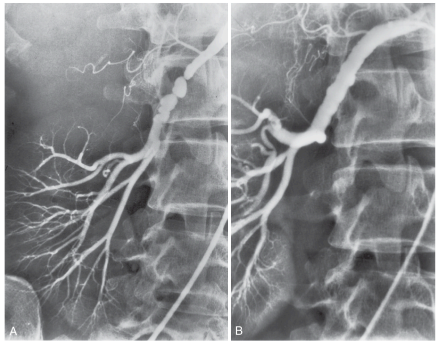


Fig. 47.8 (A and B) Angiographic appearance of medial fibroplasia with serial intravascular webs and small aneurysmal dilatations between them. These lesions appear in the midportion of the vessel, have a predilection for the right renal artery, and are most common in women. (B) As shown here, these lesions can often be improved substantially be balloon angioplasty.

Fig_47_8

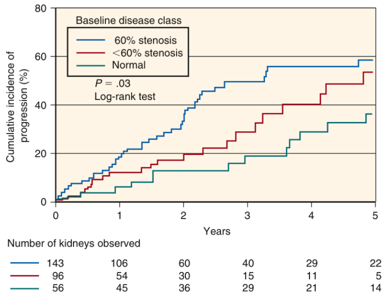


Fig. 47.9 Cumulative rates of anatomic disease progression in atherosclerotic renal artery stenosis, as measured by renal artery Doppler ultrasound. During a follow-up period of 5 years, overall progression was 31%, but those with the most severe baseline lesion progressed in 60% of cases. The progression of vascular disease was not closely related to changes in serum creatinine or renal atrophy. (Modified from Rademacher J, Weinkove R, Haller H. Techniques for predicting a favourable response to renal angioplasty in patients with renovascular disease. *Curr Opin Nephrol Hypertens*. 2001;10(6):799–805.)

Fig_47_9

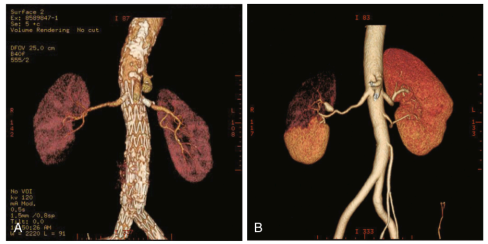


Fig. 47.10 Computed tomography (CT) angiograms illustrating reconstructed views of complex vascular disease. (A) Aortic endovascular stent extending beyond the origins of the renal arteries. Renal artery stents have been placed through the aortic graft to restore blood flow, although the nephrograms demonstrate patchy defects consistent with small vessel occlusion and/or atheroembolic events. (B) illustrates a CT angiogram with a small aneurysm of the right renal artery that has produced segmental infarction, leading to accelerated hypertension. Although CT angiography requires contrast, current multidetector CT studies allow excellent image resolution at rapid acquisition and less contrast exposure than ever before.

Fig_47_10

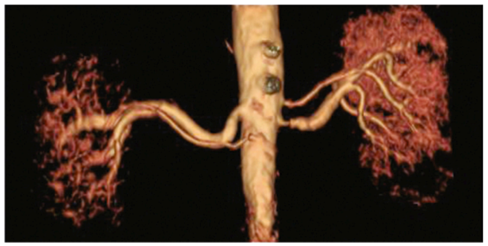


Fig. 47.11 Magnetic resonance (MR) angiogram reconstructed without gadolinium contrast. After warnings about the association of gadolinium with nephrogenic systemic fibrosis, contrast MR is less commonly used for atherosclerotic renovascular imaging than before. Newer techniques can allow excellent vascular imaging without contrast as shown here.

Fig_47_11

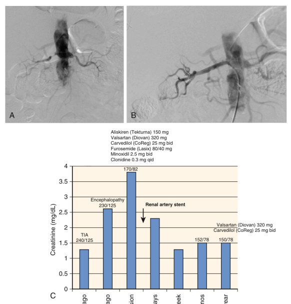


Fig. 47.12 (A) Renal aortogram illustrating high-grade, bilateral renal arterial lesions in a 73-year-old man who developed accelerated hypertension and loss of glomerular filtration rate associated with transient neurologic deficits a few months before. (B) Aortogram after placement of endovascular stents, illustrating excellent vessel patency and early technical success. This was followed by resolution of his hypertension. (C) Serum creatinine, blood pressure levels, and antihypertensive drug regimen before and after renal artery stent placement. Successful renal artery stenting led to sustained reduction in serum creatinine, improved blood pressure levels, and reduced medication requirements. Such cases reflect "high-risk" subsets that were not well represented in the prospective RCTs. (Modified figure from Textor SC. Attending rounds: a patient with accelerated hypertension and an atrophic kidney. *Clin J Am Soc Nephrol*. 2014;9(6):1117–1123.)

Fig_47_12

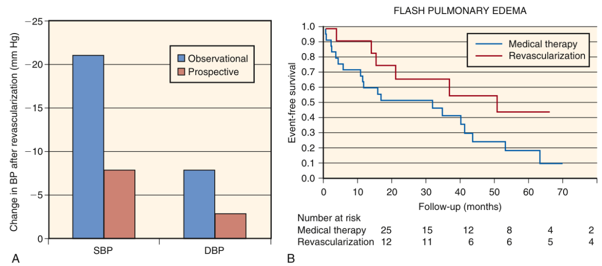


Fig. 47.13 (A and B) Comparison of blood pressure (BP) changes reported after renal revascularization from a large, observational registry (more than 1000 patients) and from a meta-analysis of three, prospective, randomized trials (210 patients). The large difference between these BP effects is reflected in variable enthusiasm for intervention between clinicians. Initial results from more recent prospective trials (ASTRAL and CORAL) identified only minor differences in achieved BP levels. While results from observational series may overstate treatment benefits, results from prospective trials likely underestimate changes, in part due to limitations in patient recruitment and crossover between treatment arms, ranging from 27% to 44% in early series (see text). (Data from[192,205] (B) Follow-up data from a registry in the United Kingdom, most of whom were not considered candidates for ASTRAL. This report identified a major mortality risk for patients with renovascular disease and episodes of pulmonary edema that was reduced for patients submitted for renal revascularization. A similar mortality benefit was identified for patients with rapidly progressive renal failure (see text). (From Ritchie and colleagues[73] with permission.)

Fig_47_13

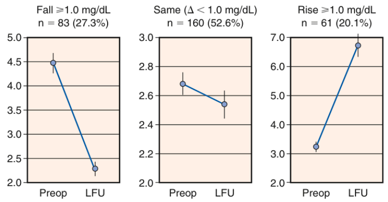


Fig. 47.14 Renal functional outcomes after renal revascularization in 304 azotemic patients (creatinine >2.0 mg/dL) with atherosclerotic renal artery stenosis (RAS). On average, mean serum creatinine does not change during a follow-up period exceeding 36 months, as observed with most reported series. Group mean values obscure major differences in clinical outcomes, as shown here. Some patients experience major clinical benefit (defined as a fall in serum creatinine of >1.0 mg/dL) (left panel). The largest group has minor changes (<1.0 mg/dL), which might be considered "stabilization" of renal function. The degree of benefit in these patients depends on whether renal function is deteriorating before intervention. The data for the group in the right panel emphasize the failure to observe consistent overall improvement in function because 18% to 22% of patients develop worsening renal function. The exact causes of this deterioration are not clear, although atherosclerotic disease is responsible for a portion. This potential hazard of revascularization must be considered when offering these procedures. *LFU,* Latest follow-up. (From Textor SC, Wilcox CS. Renal artery stenosis: a common, treatable cause of renal failure? *Annu Rev Med.* 2001;52:421–442.)

Fig_47_14

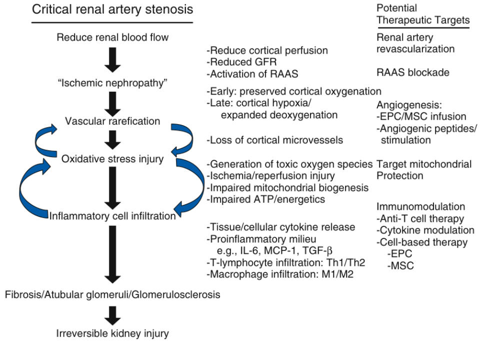


Fig. 47.15 Injury pathways and targets in renovascular disease. Working diagram highlighting recent experimental studies delineating specific pathways of oxidative stress injury and inflammatory injury pathways in the poststenotic kidney. The right panel identifies specific therapeutic targets that may alleviate these injury pathways, over and above simply restoring blood flow. Many of these pathways likely overlap in many forms of kidney injury. *EPC,* Endothelial progenitor cells; *MCP,* monocyte chemoattractant protein. (From Textor SC, Lerman LO. Paradigm shifts in atherosclerotic renovascular disease: where are we now? *J Am Soc Nephrol*. 2015;26(9):2074–2080 with permission.)

Fig_47_15

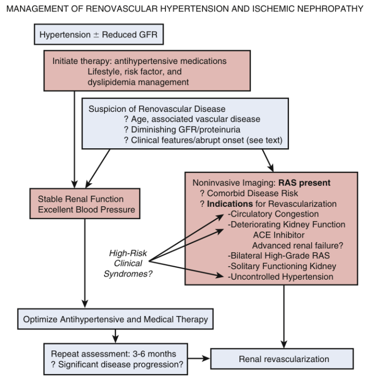
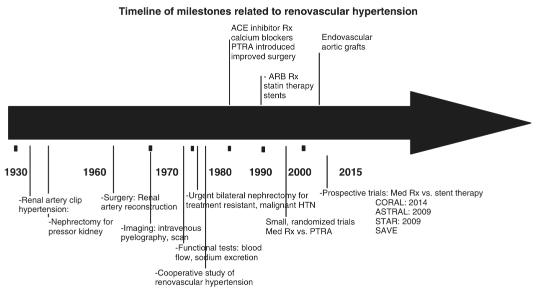


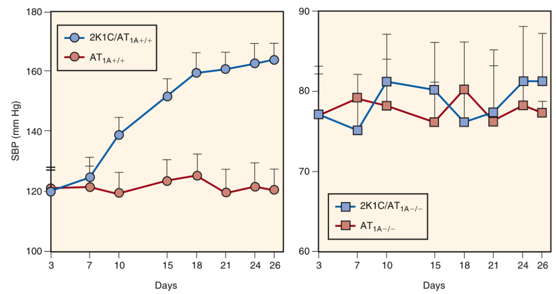
Fig. 47.16 Algorithm summarizing a management scheme for patients with renovascular hypertension and/or ischemic nephropathy. Optimizing antihypertensive and medical therapy for comorbid conditions including dyslipidemia and smoking are paramount to reducing cardiovascular morbidity and mortality in atherosclerotic disease. Decisions regarding timing of renal revascularization procedures depend on the clinical manifestations (see text) and whether blood pressure and kidney function remain stable. *ACE,* Angiotensin-converting enzyme; *GFR,* glomerular filtration rate; *PTRA,* percutaneous renal angioplasty; *RAS,* renal artery stenosis.

Fig_47_16



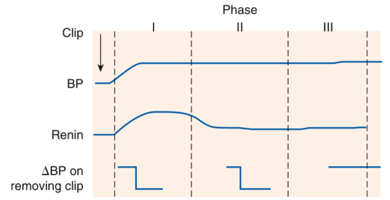
eFig. 47.1 Historical timeline of major milestones in renovascular disease (RVD). Although pressor substances derived from the kidney were identified more than a century ago, recognition that renal perfusion regulates arterial blood pressure occurred only in the 1930s. Technical advances allowing surgical reconstruction, more effective arterial imaging, and endovascular revascularization gradually evolved from 1960 to the present era. Important advances in antihypertensive drug therapy, particularly with agents that block the renin-angiotensin system, and other advances in managing atherosclerotic disease such as statins, continue to define optimal medical management of patients with RVD. The introduction of endovascular aortic stent grafts represents an iatrogenic form of RVD that benefits from protection of the renal artery patency. Results from several prospective, randomized clinical trials indicate that renal revascularization often fails to add substantial benefits to effective medical therapy in the near term for patients with moderate atherosclerotic RVD, although high-risk subsets are an exception. *ACE,* Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker; *PTRA,* percutaneous transluminal renal angioplasty.

Fig_e47_1



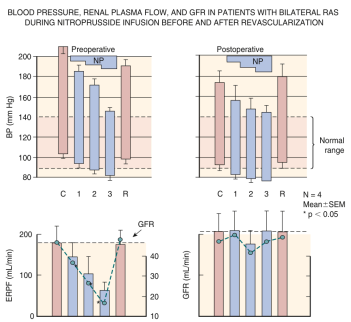
eFig. 47.2 Systolic blood pressures in mice before and after placement of a renal artery clip in experimental two-kidney, one-clip renovascular hypertension (2K1C). The rise in systolic blood pressure *(SBP)* after clip placement develops rapidly only in mice with an intact angiotensin 1A receptor ($AT_{1A+/+}$; *blue circles, left panel*). This rise is blocked by administration of an angiotensin receptor blocker *(red circles, left panel).* A genetic knockout mouse strain with no AT1A receptor ($AT_{1A-/-}$) has a lower SBP and no change after renal artery clipping *(blue squares, right panel).* No additional effect is noted with an angiotensin receptor blocker *(red squares, right panel).* These data reinforce the essential role of the renin-angiotensin system and an intact angiotensin 1 receptor for the development of renovascular hypertension. (Modified from Cervenka L, Horacek V, Vaneckova I, et al. Essential role of AT1-A receptor in the development of 2K1C hypertension. *Hypertension.* 2002;40:735–741.)

Fig_e47_2



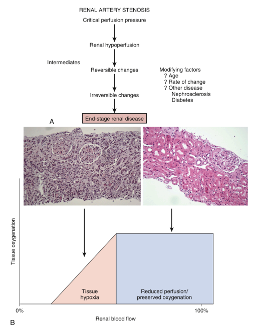
eFig. 47.3 Schematic depiction of phases observed in experimental renovascular hypertension. Initially, high levels of renin activity fall in the chronic phase, although removal of the renal artery clip corrects hypertension. These observations support the concept of renal artery stenosis leading to the recruitment of additional structural and pressor mechanisms after initial activation of the renin-angiotensin system. The degree to which human renovascular hypertension follows these patterns is not well known. *BP,* Blood pressure. (From Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector's: The Kidney.* 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

Fig_e47_3



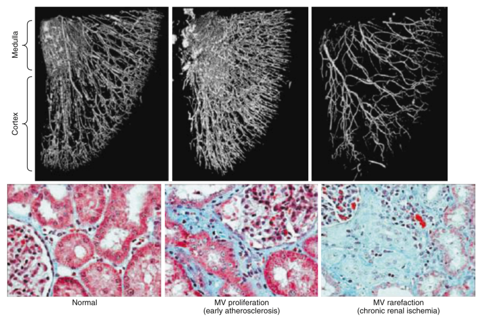
eFig. 47.4 Effective renal plasma flow (ERPF) and glomerular filtration rate (GFR) in patients with critical bilateral renal artery stenosis during pressure reduction with sodium nitroprusside (NP). Reducing systemic blood pressure to normal levels produced a reversible fall in both plasma flow and GFR. Repeat studies in the same patients (right-hand panel) after unilateral surgical revascularization demonstrate that the sensitivity of blood flow and GFR to pressure reduction can be reversed. *SEM,* Standard error of the mean. (From Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector's: The Kidney.* 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

Fig_e47_4



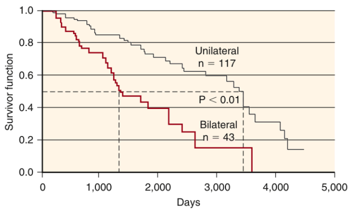
eFig. 47.5 (A) General clinical paradigm that a reduction in the renal blood flow related to hypoperfusion of the kidney can be reversible but can undergo a transition to irreversible injury eventually that no longer responds to restoration of blood flow alone. (B) Schematic depiction of the adaptation of tissue oxygenation in the kidney to moderate reductions in blood flow. Lowering renal blood flow by up to 30% to 40% in human subjects (right side) is associated with preservation of both oxygen gradients (lower graph) and tissue histology (upper images). The fact that such adaptation occurs partly explains the stability of kidney function during antihypertensive drug therapy that lowers systemic pressure and renal blood flow in patients with atherosclerotic renal artery stenosis. Eventually, more severe reductions of blood flow overwhelm "adaptation," leading to overt tissue hypoxia (left side). (Magnification: low power: ×40.) (Modified from Gloviczki ML, Keddis MT, Garovic VD, et al. TGF expression and macrophage accumulation in atherosclerotic renal artery stenosis. *Clin J Am Soc Nephrol.* 2013;8(4):546–553; and Textor SC, Lerman LO. Paradigm shifts in atherosclerotic renovascular disease: where are we now? *J Am Soc Nephrol.* 2015;26:2074–2080.)

Fig_e47_5



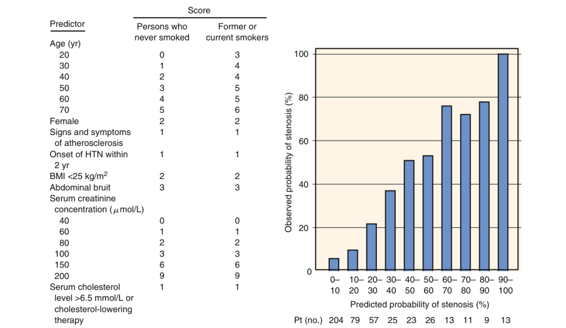
eFig. 47.6 *Upper panels,* Microcomputed tomography reconstructions of vessels in experimental atherosclerosis and superimposed large-vessel renovascular disease at an early stage (middle) and after chronic ischemia (right), compared with normal (left). *Lower panels,* Corresponding renal histology (trichrome stain ×400). Complex microvascular dysfunction and rarefaction develop in poststenotic kidneys. These are accelerated and/or modified by angiogenic stimuli, oxidative stress pathways, and a variety of cytokines, leading eventually to interstitial fibrosis. Studies in experimental animals suggest that angiogenic stimuli such as intrarenal infusion of endothelial progenitor cells or mesenchymal stem cells may offer the potential to repair microvascular injury. *MV,* Microvascular. (From Lerman LO, Chade AR. Angiogenesis in the kidney: a new therapeutic target? *Curr Opin Nephrol Hyper.* 2009;18:160–165.)

Fig_e47_6



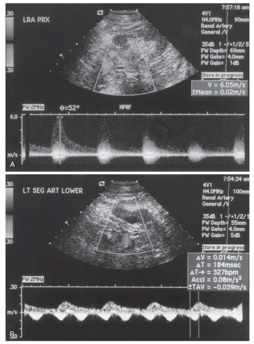
eFig. 47.7 Kaplan-Meier survival curve of 160 patients with more than 70% renal artery stenosis managed without revascularization. Those with bilateral disease had lower survival, primarily due to associated cardiovascular disease. The mean age of death was 79 years. These data underscore the close relationship between the extent of vascular disease and mortality. Less than 10% of these subjects developed advanced kidney disease during follow-up, although long-term survival, even in treated patients, is related to levels of kidney function at the time of intervention. Recent prospective randomized clinical trials indicate that between 16% and 22% of medically treated patients with atherosclerotic renal artery stenosis progress to more advanced renal dysfunction over 3 to 5 years. (From Foster JH, Maxwell MH, Franklin SS, et al. Renovascular occlusive disease: results of operative treatment. *JAMA.* 975;2231:1043–1198; Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's: The Kidney.* 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

Fig_e47_7



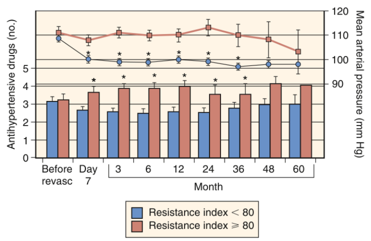
eFig. 47.8 Probability of identifying renal artery stenosis based on clinical features. These data were derived from 477 patients in referral centers for treatment-resistant hypertension *(HTN)* in the Netherlands. Overall prevalence was 22.4%, illustrating that even in "enriched" patient populations, renovascular disease is not present in the majority. Clinical features allowed selection of patients for testing with a relatively high pretest probability of disease, which affects the validity of testing schemes. (From Krijnen P, van Jaarsveld BC, Steyerberg EW, et al. A clinical prediction rule for renal artery stenosis. Ann Intern Med. 1998;129:705–711.)

Fig_e47_8



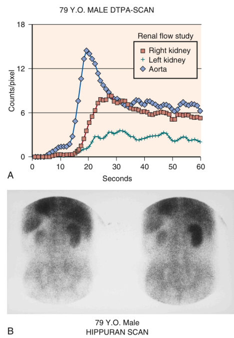
eFig. 47.9 (A) Duplex ultrasound velocity measurement in a patient with high-grade renal artery stenosis affecting the proximal left renal artery *(LRA PRX)*. Peak systolic velocities reach 605 cm/s (6.05 m/s), well above the normal upper limit of 180 cm/s. (B) Segmental arterial branch ultrasound in the distal segmental renal arteries demonstrates "parvus" and "tardus" dampening of the signal characteristic of poststenotic waveforms. The utility of these measurements depends on the ability to obtain reliable identification of vessel segments and the skills of the operator. Once the location of a vascular lesion is known, subsequent studies can be performed more easily to track progression of vascular occlusion, restenosis, and/or the results of endovascular intervention.

Fig_e47_9



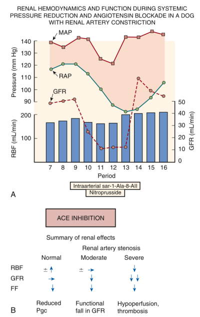
eFig. 47.10 Outcome of revascularization as measured by mean arterial blood pressure and number of antihypertensive agents in 138 patients with renal artery stenosis. These patients were divided into groups with ultrasound-determined resistive index above 80 and those lower than 80 in the most severely affected kidney. The authors indicate that a high resistive index reflects intrinsic parenchymal and small vessel disease in the kidney that does not improve after revascularization. Those with lower indices had both lower blood pressures during follow-up and lower antihypertensive medication requirements. (From Radermacher J, Chavan A, Bleck J, et al. Use of Doppler ultrasonography to predict the outcome of therapy for renal-artery stenosis. *N Engl J Med.* 2001;344:410–417.)

Fig_e47_10



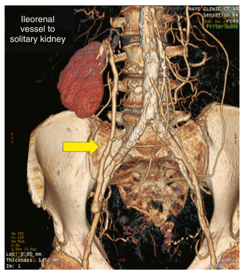
eFig. 47.11 (A and B) Isotope renography in a patient with unilateral renal artery stenosis. (A) A ^{99m}Tc-DTPA scan demonstrates delayed circulation and excretion of isotope on the *left*. (B) A hippuran scan (now replaced with ^{99m}Tc-mercaptoacetyltriglycine) provides a renogram demonstrating a small kidney with impaired renal function on the affected side. Radionuclide scans provide a comparative estimate of function from each kidney that may facilitate selection of intervention including the potential effect of nephrectomy. *DTPA,* Diethylenetriamine pentaacetic acid.

Fig_e47_11



eFig. 47.12 (A) The glomerular filtration rate *(GFR)* falls beyond a renal artery stenotic lesion during blockade of angiotensin II (produced by intrarenal infusion of sar-1-ala-8-angiotensin II) and pressure reduction induced by sodium nitroprusside. The fall in GFR occurs despite preserved renal blood flow *(RBF)*, measured by electromagnetic flow probe. These observations illustrate the specific role of angiotensin II in maintaining GFR in the poststenotic kidney at reduced perfusion pressures. (B) Summary of the effects of angiotensin-converting enzyme *(ACE)* inhibition and/or angiotensin receptor blockade on RBF, GFR, and filtration fraction *(FF)* in normal, moderate, and severe levels of renal artery stenosis. As compared with other antihypertensive agents, ACE inhibitors and angiotensin receptor blockers (ARBs) lead to a fall in GFR and filtration fraction due to removal of the efferent arteriolar effects of angiotensin II. When stenosis is sufficiently severe that pressure reduction compromises renal blood flow, the potential for complete occlusion is present, as with other effective antihypertensive agents as well. *MAP,* Mean arterial pressure; *RAP,* renal artery pressure. (From Textor SC. Renal failure related to ACE inhibitors. *Semin Nephrol.* 1997;17:67–76.)

Fig_e47_12



eFig. 47.13 Algorithm summarizing a management scheme for patients with renovascular hypertension and/or ischemic nephropathy. Optimizing antihypertensive and medical therapy for comorbid conditions including dyslipidemia and smoking is paramount to reducing cardiovascular morbidity and mortality in atherosclerotic disease. Decisions regarding the timing of renal revascularization procedures depend on the clinical manifestations (see text) and whether blood pressure and kidney function remain stable. *ACE,* Angiotensin-converting enzyme; *GFR,* glomerular filtration rate; *PTRA,* percutaneous renal angioplasty; *RAS,* renal artery stenosis.

Fig_e47_13

Table 47.1 Prevalence rates of atherosclerotic renal artery stenosis[a]

Vascular disease	Prevalence rate (%)
Suspected renovascular hypertension	14.1
Coronary angiography	10.5
With hypertension	17.8
Peripheral vascular disease	25.3
AAA	33.1
ESKD	40.8 (?)[b]
Congestive heart failure	54.1 (?)[b]

[a]40 studies, 15,879 patients. Shown are prevalence rates of patients with vascular disease affecting other regional beds identified by angiography.

[b](?) marks refer to limited data suggesting high prevalence in unexplained advanced chronic kidney disease (CKD) in older subjects and patients with congestive heart failure.

AAA, Abdominal aortic aneurysm; *ESKD,* end-stage kidney disease.

Modified from de Mast Q, Beutler JJ. The prevalence of atherosclerotic renal artery stenosis in risk groups: a systematic literature review. *J Hypertens*. 2009;27: 1333–1240.

Table_47_1

Table 47.2 Clinical features of patients with renovascular hypertension[a]

Clinical feature	Essential HTN (%)	Renovascular HTN (%)
Duration <1 year	12	24
Age of onset >50 years	9	15
Family history of HTN	71	46
Grade 3 or 4 fundi	7	15
Abdominal bruit	9	46
Blood urea nitrogen > 20 mg/dL	8	16
Potassium <3.4 mEq/L	8	16
Urinary casts	9	20
Proteinuria	32	46

[a]Clinical features that differed ($P < .05$) between closely matched groups of 131 patients with essential versus renovascular hypertension taken from the Cooperative Study of Renovascular Hypertension in the 1960s.

HTN, Hypertension.

(From Simon N, Franklin SS, Bleifer KH, Maxwell MH. Clinical characteristics of renovascular hypertension. *JAMA*. 1972;220:1209–1218.) These observations underscore the potential severity of hypertension in candidates for surgery, but none of these features allows clinical discrimination with confidence.

Table_47_2

Table 47.3 Noninvasive assessment of renal artery stenosis

Study	Rationale	Strengths	Limitations
Physiologic studies to assess the renin-angiotensin system			
Measurement of peripheral plasma renin activity	Reflects the level of sodium excretion	Measures the level of activation of the renin-angiotensin system	Low predictive accuracy for renovascular hypertension; results influenced by medications and many other conditions
Measurement of captopril-stimulated renin activity	Produces a fall in pressure distal to the stenosis	Enhances the release of renin from the stenotic kidney	Low predictive accuracy for renovascular hypertension; results influenced by many other conditions
Measurement of renal vein renin activity	Compares renin release from the two kidneys	Lateralization predictive of improvement in blood pressure with revascularization	Nonlateralization has limited predictive power of the failure of blood pressure to improve after revascularization; results influenced by medications and many other conditions
Perfusion studies to assess differential renal blood flow			
Captopril renography with technetium ^{99m}Tc mertiatide (^{99m}Tc MAG3)	Captopril-mediated fall in filtration pressure amplifies differences in renal perfusion	Normal study excludes renovascular hypertension	Multiple limitations in patients with advanced atherosclerosis or creatinine >2.0 mg/dL (177 µmol/L)
Nuclear imaging with technetium mertiatide or technetium-labeled pentetic acid (DTPA) to estimate fractional flow to each kidney	Estimates fractional flow to each kidney	Allows calculation of single-kidney glomerular filtration rate	Results may be influenced by other conditions (e.g., obstructive uropathy)
Vascular studies to evaluate the renal arteries			
Duplex ultrasonography	Shows the renal arteries and measures flow velocity as a means of assessing the severity of stenosis	Inexpensive; widely available: suitable for sequential measurement to follow disease progression and/or restenosis	Heavily dependent on operator's experience; less useful than invasive angiography for the diagnosis of fibromuscular dysplasia and abnormalities in accessory renal arteries
Magnetic resonance angiography	Shows the renal arteries and perirenal aorta	Not nephrotoxic, but concerns for gadolinium toxicity exclude use in GFR <30 mL/min/1.73 m^2; provides excellent images	Expensive; gadolinium excluded in renal failure, unable to visualize stented vessels
Computed tomographic angiography	Shows the renal arteries and perirenal aorta	Provides excellent images; stents do not cause artifacts	Expensive, moderate volume of contrast required, potentially nephrotoxic

Modified from Safian RD, Textor SC. Medical progress: renal artery stenosis. *N Engl J Med*. 2001;344:431–442 with permission.

the role of RVD in explaining deteriorating renal function? The specific approach to diagnosis will differ depending on which of these is the predominant clinical objective.

a discussion of some of the specific merits and limitations of each modality as they apply to application in renovascular hypertension and ischemic nephropathy[96] (Table 47.3).

Table_47_3

Table 47.4 Prospective, randomized clinical trials for renal function and/or cardiovascular outcomes comparing medical therapy versus renal artery stenting

Trials	Number	Population	Inclusion criteria	Exclusion criteria	Outcomes
STAR (2009) 10 centers f/up 2 years	Med Tx:76 PTRA: 64	Patients with impaired renal function, ostial ARVD detected by various imaging studies and stable blood pressure on statin and aspirin	ARVD >50% Creatinine clearance <80 mL/min/1.73 m^2 Controlled blood pressure 1 month before inclusion	Kidney <8 cm and renal artery diameter <4 mm, eCrCl <15 mL/min per 1.73 m^2, DM with proteinuria (>3 g/d), malignant hypertension	No difference in GFR decline (primary endpoint ≥20% change in clearance), but many did not undergo PTRA due ARVD <50% on angiography Serious complications in the PTRA group Study was underpowered
ASTRAL (2009) 57 centers f /up 5 years	Med Tx:403 PTRA: 403	Patients with uncontrolled or refractory hypertension or unexplained renal dysfunction with unilateral or bilateral ARVD on statin and aspirin	ARVD substantial disease suitable for endovascular and patient's doctor uncertainty of clinical benefit from revascularization	High likelihood of PTRA in <6 mo Without ARVD, previous ARVD PTRA FMD	No difference in BP, renal function, mortality, CV events (primary endpoint: 20% reduction in the mean slope of the reciprocal of the serum creatinine level) Substantial risk in the PTRA group
CORAL (2013) 109 centers f/up 5 years	Med Tx:480 PTRA:467	Hypertension 2 or more antihypertensives or CKD stage ≥3 with ARVD with unilateral or bilateral disease on statin	SBP >155 mm Hg, at least 2 drugs ARVD >60% Subsequent changes included that the SBP >155 mm Hg for defining systolic hypertension was no longer specified as long as patient had CKD stage 3	FMD Creatinine >4.0 mg/dL kidney length <7 cm and use of >1 stent	No difference in death from CV or renal causes. Modest improvement of SBP in the stented group Total 26 complications (5.5%)

ARVD, Atherosclerotic renovascular disease; *CKD,* chronic kidney disease; *CV,* cardiovascular; *eClCr,* estimated creatinin clearance; *FMD,* fibromuscular dysplasia; *PTRA,* percutaneous renal artery angioplasty; *RVD,* renovascular disease; *SBP,* systolic blood pressure; *Tx,* therapy.
(Modified from Bhalla V, Textor SC, Beckman JA, et al. Revascularization for renovascular disease: a scientific statement from the American Heart Association. *Hypertension*. 2022;79(8):e128–e143.

Table_47_4

Table 47.5 Clinical scenarios in which treatment of significant renal artery stenosis may be considered[a]

Type of care	Examples
Appropriate care	• Cardiac disturbance syndromes (flash pulmonary edema or acute coronary syndrome [ACS]) with severe hypertension • Resistant hypertension (HTN; uncontrolled with failure of maximally tolerated doses of at least 3 antihypertensive agents, 1 of which is a diuretic, or intolerance to medications) • Ischemic nephropathy with chronic kidney disease (CKD), with eGFR <45 mL/min/1.73 m^2 and global renal ischemia (unilateral significant RAS with a solitary kidney or bilateral significant RAS) without other explanation
May be appropriate care	• Unilateral RAS with CKD (eGFR <45 mL/min/1.73 m^2) • Unilateral RAS with prior episodes of congestive heart failure (stage C) • Anatomically challenging or high-risk lesion (early bifurcation, small vessel, severe concentric calcification, severe aortic atheroma or mural thrombus)
Rarely appropriate care	• Unilateral, solitary, or bilateral RAS with controlled BP and normal renal function. • Unilateral, solitary, or bilateral RAS with kidney size <7 cm in pole-to-pole length • Unilateral, solitary, or bilateral RAS with chronic end-stage kidney disease on hemodialysis >3 months • Unilateral, solitary, or bilateral renal artery chronic total occlusion

[a]Significant RAS is an angiographically moderate lesion (50%–70%) with physiologic confirmation of severity or >70% stenosis.
BP, Blood pressure; *eGFR,* estimated glomerular filtration rate.
Modified from Parikh SA, Shishehbor MH, Gray BH, et al. SCAI expert consensus statement for renal artery stenting appropriate use. *Catheter Cardiovasc Intervent*. 2014;84:1163–1171.

Table_47_5

eTable 47.1 Summary of observational series in renal revascularization

T1 outcomes of renal artery angioplasty, ±stent placement; hypertension			
	Cured[a]	**Improved**	**No Change**
14 series, 678 patients, 98% technical success	Weighted mean, 17%; range, 3%–68%	Weighted mean, 47%; range, 5%–61%	Weighted mean, 36% range, 0%–61%
Renovascular hypertension, 472 patients,	12%	73%	15%
T1 renal artery stents: effect on renal function[b] in azotemic patients			
	"Improved"	**"Stabilized"**	**Worse**
14 series; reporting "impaired renal function"; 496 patients	Weighted mean, 30%; range, 10%–41%	Weighted mean, 42%; range, 32%–71%	Weighted mean, 29%; range, 19%–34%
Ischemic nephropathy; 469 patients	41%	37%; no change	22%

[a]Cured, blood pressure at normal levels, no medication; improved, blood pressure lower by at least 10 mm Hg, fewer medications; no change, same blood pressure and/or same medications.
[b]Renal function—variably defined as improved, persistent fall in creatinine, variable from 0.3 to 1.0 mg/dL or more; worse, variable, but persistent increase in creatinine 0.3 to 1.0 mg/dL or more.
Modified from Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's: The Kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.

Table_e47_1

eTable 47.2 Prospective randomized trials of medical therapy versus angioplasty for hypertension[a]

Source	Inclusion, blood pressure measurement	Blood pressure outcome (mm Hg)	Renal outcome (mm Hg)	Comments
SNRAS, Webster et al., 1998[127]; n = 55 (unilateral = 27); n = 135 eligible; RAS > 50%	DBP ≥95 mm Hg, 2 drugs; exclusion: CVA, MI within 3 mo; creatinine >500 µmol/L; BP, random-zero device; no ACE inhibitors allowed	Unilateral—PTRA, 173/95 Med Rx, 161/88 Bilateral—PTRA, 152/83; Med Rx, 171/91; P < .01	Creatinine (µmol/L), bilateral PTRA,188 Med Rx, 157 Unilateral PTRA, 144 Med Rx, 168	"… unable to demonstrate any benefit in respect of renal function or event-free survival" (F/U 40 mo)
EMMA: Plouin et al., 1998[240]; n = 49 (unilateral RAS only); RAS >75% or >60%, lateralizing study	Age <75 years; normal contralateral kidney; exclusion: malignant HTN CVA, CHF, MI within 6 mo; BP, automated sphygmomanometer, ABPM at 6 mo	PTRA, 140/81 Med Rx, 141/84 No. drugs (DDD)—PTRA, 1.0 Med Rx, 1.78, P < .01 Crossover to PTRA: 7/26 (27%)	Creatinine Clearance mL/min: (6 mo) PTRA: 77 Med Rx: 74 Renal artery occlusion: PTRA: 0 Med Rx: 0	"BP levels and the proportion of patients given antihypertensive treatment were similar 1 year after randomization in the control and angioplasty groups, confirming that the BP-lowering effect of angioplasty in the short and medium terms is limited in atherosclerotic RAS"
DRASTIC: Van Jaarsveld, et al., 2000[133]; n = 106 ASO RAS >50%	Resistant: 2 drugs; DBP >95 mm Hg or creatinine rise with ACEI; exclusion: creatinine ≥2.3 mg/dL; solitary kidney, total occlusion—kidney <8 cm; BP, automated oscillometric	BP outcomes at 3 mo—PTRA, 169/89 Med Rx, 163/88 at 12 mo—PTRA,152/84 Med Rx: 162/88 No. of drugs, 1.9 vs. 2.4, P < .01	Creatinine Cl (3 mo) mL/min: PTRA: 70 Med Rx: 59 (P = .03) Abnormal renograms PTRA: 36% Med Rx: 70% (P = .002) Renal artery occlusion PTRA 0 Med Rx 8	"In the treatment of patients with hypertension and renal artery stenosis, angioplasty has little advantage over antihypertensive drug therapy."

[a]In patients with atherosclerotic renal artery stenosis.
ABPM, Overnight ambulatory reading; *ACEI,* angiotensin-converting enzyme inhibitor; *ASO,* atherosclerotic occlusive disease; *BP,* blood pressure; *CHF,* congestive heart failure; *CVA,* cerebrovascular accident; *DBP,* diastolic blood pressure; *DDD,* defined daily doses; *DRASTIC,* Dutch Renal Artery Stenosis Intervention Cooperative; *EMMA,* Essai Multicentrique Medicaments vs. Angioplastie; *F/U,* follow-up; *HTN,* hypertension; *Med Rx,* medical therapy; *MI,* myocardial infarction; *n,* number of patients; *PTRA,* percutaneous renal artery angioplasty; *RAS,* renal artery stenosis; *SNRAS,* Scottish and Newcastle Renal Artery Stenosis Collaborative Group; *Tx,* therapy.

Table_e47_2